MSG No. 166.99 $R\bar{3}'m$ [Type III, trigonal]

Table 1: Wyckoff site: `3a`, site symmetry: `-3'm`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`

Table 2: Wyckoff site: `3b`, site symmetry: `-3'm`.

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`

Table 3: Wyckoff site: `6c`, site symmetry: `3m`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$[0,\ 0,\ -z]$	`[7,8,9,10,11,12]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$	`[13,14,15,16,17,18]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}-z\right]$	`[19,20,21,22,23,24]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$	`[25,26,27,28,29,30]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}-z\right]$	`[31,32,33,34,35,36]`

Table 4: Wyckoff site: `9d`, site symmetry: `.2'/m`.

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,4,7,10]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,9,12]`
4	$\left[\frac{1}{6},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[13,16,19,22]`
5	$\left[\frac{2}{3},\ \frac{5}{6},\ \frac{5}{6}\right]$	`[14,17,20,23]`
6	$\left[\frac{1}{6},\ \frac{5}{6},\ \frac{5}{6}\right]$	`[15,18,21,24]`
7	$\left[\frac{5}{6},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[25,28,31,34]`
8	$\left[\frac{1}{3},\ \frac{1}{6},\ \frac{1}{6}\right]$	`[26,29,32,35]`
9	$\left[\frac{5}{6},\ \frac{1}{6},\ \frac{1}{6}\right]$	`[27,30,33,36]`

Table 5: Wyckoff site: `9e`, site symmetry: `.2'/m`.

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,7,10]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,6,9,12]`

continued ...

Table 5

No.	position	mapping
4	$\left[\frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right]$	`[13,16,19,22]`
5	$\left[\frac{2}{3}, \frac{5}{6}, \frac{1}{3}\right]$	`[14,17,20,23]`
6	$\left[\frac{1}{6}, \frac{5}{6}, \frac{1}{3}\right]$	`[15,18,21,24]`
7	$\left[\frac{5}{6}, \frac{2}{3}, \frac{2}{3}\right]$	`[25,28,31,34]`
8	$\left[\frac{1}{3}, \frac{1}{6}, \frac{2}{3}\right]$	`[26,29,32,35]`
9	$\left[\frac{5}{6}, \frac{1}{6}, \frac{2}{3}\right]$	`[27,30,33,36]`

Table 6: Wyckoff site: `18f`, site symmetry: `.2'.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,7]`
2	$[0, x, 0]$	`[2,8]`
3	$[-x, -x, 0]$	`[3,9]`
4	$[-x, 0, 0]$	`[4,10]`
5	$[0, -x, 0]$	`[5,11]`
6	$[x, x, 0]$	`[6,12]`
7	$\left[x + \frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right]$	`[13,19]`
8	$\left[\frac{2}{3}, x + \frac{1}{3}, \frac{1}{3}\right]$	`[14,20]`
9	$\left[\frac{2}{3} - x, \frac{1}{3} - x, \frac{1}{3}\right]$	`[15,21]`
10	$\left[\frac{2}{3} - x, \frac{1}{3}, \frac{1}{3}\right]$	`[16,22]`
11	$\left[\frac{2}{3}, \frac{1}{3} - x, \frac{1}{3}\right]$	`[17,23]`
12	$\left[x + \frac{2}{3}, x + \frac{1}{3}, \frac{1}{3}\right]$	`[18,24]`
13	$\left[x + \frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$	`[25,31]`
14	$\left[\frac{1}{3}, x + \frac{2}{3}, \frac{2}{3}\right]$	`[26,32]`
15	$\left[\frac{1}{3} - x, \frac{2}{3} - x, \frac{2}{3}\right]$	`[27,33]`
16	$\left[\frac{1}{3} - x, \frac{2}{3}, \frac{2}{3}\right]$	`[28,34]`
17	$\left[\frac{1}{3}, \frac{2}{3} - x, \frac{2}{3}\right]$	`[29,35]`
18	$\left[x + \frac{1}{3}, x + \frac{2}{3}, \frac{2}{3}\right]$	`[30,36]`

Table 7: Wyckoff site: `18g`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,7]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,8]`
3	$\left[-x, -x, \frac{1}{2}\right]$	`[3,9]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[4,10]`
5	$\left[0, -x, \frac{1}{2}\right]$	`[5,11]`
6	$\left[x, x, \frac{1}{2}\right]$	`[6,12]`
7	$\left[x + \frac{2}{3}, \frac{1}{3}, \frac{5}{6}\right]$	`[13,19]`
8	$\left[\frac{2}{3}, x + \frac{1}{3}, \frac{5}{6}\right]$	`[14,20]`
9	$\left[\frac{2}{3} - x, \frac{1}{3} - x, \frac{5}{6}\right]$	`[15,21]`
10	$\left[\frac{2}{3} - x, \frac{1}{3}, \frac{5}{6}\right]$	`[16,22]`
11	$\left[\frac{2}{3}, \frac{1}{3} - x, \frac{5}{6}\right]$	`[17,23]`
12	$\left[x + \frac{2}{3}, x + \frac{1}{3}, \frac{5}{6}\right]$	`[18,24]`

continued ...

Table 7

No.	position	mapping
13	$\left[x+\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$	[25,31]
14	$\left[\frac{1}{3},\ x+\frac{2}{3},\ \frac{1}{6}\right]$	[26,32]
15	$\left[\frac{1}{3}-x,\ \frac{2}{3}-x,\ \frac{1}{6}\right]$	[27,33]
16	$\left[\frac{1}{3}-x,\ \frac{2}{3},\ \frac{1}{6}\right]$	[28,34]
17	$\left[\frac{1}{3},\ \frac{2}{3}-x,\ \frac{1}{6}\right]$	[29,35]
18	$\left[x+\frac{1}{3},\ x+\frac{2}{3},\ \frac{1}{6}\right]$	[30,36]

Table 8: Wyckoff site: 18h, site symmetry: .m.

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,5]
2	$[x,\ 2x,\ z]$	[2,6]
3	$[-2x,\ -x,\ z]$	[3,4]
4	$[2x,\ x,\ -z]$	[7,12]
5	$[-x,\ x,\ -z]$	[8,10]
6	$[-x,\ -2x,\ -z]$	[9,11]
7	$\left[x+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	[13,17]
8	$\left[x+\frac{2}{3},\ 2x+\frac{1}{3},\ z+\frac{1}{3}\right]$	[14,18]
9	$\left[\frac{2}{3}-2x,\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	[15,16]
10	$\left[2x+\frac{2}{3},\ x+\frac{1}{3},\ \frac{1}{3}-z\right]$	[19,24]
11	$\left[\frac{2}{3}-x,\ x+\frac{1}{3},\ \frac{1}{3}-z\right]$	[20,22]
12	$\left[\frac{2}{3}-x,\ \frac{1}{3}-2x,\ \frac{1}{3}-z\right]$	[21,23]
13	$\left[x+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	[25,29]
14	$\left[x+\frac{1}{3},\ 2x+\frac{2}{3},\ z+\frac{2}{3}\right]$	[26,30]
15	$\left[\frac{1}{3}-2x,\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	[27,28]
16	$\left[2x+\frac{1}{3},\ x+\frac{2}{3},\ \frac{2}{3}-z\right]$	[31,36]
17	$\left[\frac{1}{3}-x,\ x+\frac{2}{3},\ \frac{2}{3}-z\right]$	[32,34]
18	$\left[\frac{1}{3}-x,\ \frac{2}{3}-2x,\ \frac{2}{3}-z\right]$	[33,35]

Table 9: Wyckoff site: 36i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$[-x+y,\ y,\ z]$	[4]
5	$[-y,\ -x,\ z]$	[5]
6	$[x,\ x-y,\ z]$	[6]
7	$[x-y,\ -y,\ -z]$	[7]
8	$[y,\ x,\ -z]$	[8]
9	$[-x,\ -x+y,\ -z]$	[9]
10	$[-x,\ -y,\ -z]$	[10]
11	$[y,\ -x+y,\ -z]$	[11]
12	$[x-y,\ x,\ -z]$	[12]

continued ...

Table 9

No.	position	mapping
13	$\left[x+\frac{2}{3}, y+\frac{1}{3}, z+\frac{1}{3}\right]$	[13]
14	$\left[\frac{2}{3}-y, x-y+\frac{1}{3}, z+\frac{1}{3}\right]$	[14]
15	$\left[-x+y+\frac{2}{3}, \frac{1}{3}-x, z+\frac{1}{3}\right]$	[15]
16	$\left[-x+y+\frac{2}{3}, y+\frac{1}{3}, z+\frac{1}{3}\right]$	[16]
17	$\left[\frac{2}{3}-y, \frac{1}{3}-x, z+\frac{1}{3}\right]$	[17]
18	$\left[x+\frac{2}{3}, x-y+\frac{1}{3}, z+\frac{1}{3}\right]$	[18]
19	$\left[x-y+\frac{2}{3}, \frac{1}{3}-y, \frac{1}{3}-z\right]$	[19]
20	$\left[y+\frac{2}{3}, x+\frac{1}{3}, \frac{1}{3}-z\right]$	[20]
21	$\left[\frac{2}{3}-x, -x+y+\frac{1}{3}, \frac{1}{3}-z\right]$	[21]
22	$\left[\frac{2}{3}-x, \frac{1}{3}-y, \frac{1}{3}-z\right]$	[22]
23	$\left[y+\frac{2}{3}, -x+y+\frac{1}{3}, \frac{1}{3}-z\right]$	[23]
24	$\left[x-y+\frac{2}{3}, x+\frac{1}{3}, \frac{1}{3}-z\right]$	[24]
25	$\left[x+\frac{1}{3}, y+\frac{2}{3}, z+\frac{2}{3}\right]$	[25]
26	$\left[\frac{1}{3}-y, x-y+\frac{2}{3}, z+\frac{2}{3}\right]$	[26]
27	$\left[-x+y+\frac{1}{3}, \frac{2}{3}-x, z+\frac{2}{3}\right]$	[27]
28	$\left[-x+y+\frac{1}{3}, y+\frac{2}{3}, z+\frac{2}{3}\right]$	[28]
29	$\left[\frac{1}{3}-y, \frac{2}{3}-x, z+\frac{2}{3}\right]$	[29]
30	$\left[x+\frac{1}{3}, x-y+\frac{2}{3}, z+\frac{2}{3}\right]$	[30]
31	$\left[x-y+\frac{1}{3}, \frac{2}{3}-y, \frac{2}{3}-z\right]$	[31]
32	$\left[y+\frac{1}{3}, x+\frac{2}{3}, \frac{2}{3}-z\right]$	[32]
33	$\left[\frac{1}{3}-x, -x+y+\frac{2}{3}, \frac{2}{3}-z\right]$	[33]
34	$\left[\frac{1}{3}-x, \frac{2}{3}-y, \frac{2}{3}-z\right]$	[34]
35	$\left[y+\frac{1}{3}, -x+y+\frac{2}{3}, \frac{2}{3}-z\right]$	[35]
36	$\left[x-y+\frac{1}{3}, x+\frac{2}{3}, \frac{2}{3}-z\right]$	[36]